General Mathematics — Exercise Sheet 32

Motion along a path

Variant: Statements

Exercise 1 (Reconstruction with a changing acceleration). A motion $\boldsymbol{r} : \mathbb{R} \to \mathbb{R}^3$ has acceleration $\boldsymbol{a}(t) = (6t, -4, 2\cos t)$, initial position $\boldsymbol{r}(0) = (2, -1, 0)$ and initial velocity $\boldsymbol{v}(0) = (1, 3, -2)$. Use Lecture 32's reconstruction integrals to find $\boldsymbol{v}$ and $\boldsymbol{r}$, and prove that the resulting C^2 motion is unique on $\mathbb{R}$. Use the scalar primitives of Lecture 23 and the fundamental theorem of Lecture 26.

Exercise 2 (An expanding or contracting spiral). Let $a > 0$, $b \in \mathbb{R}$ and $\omega > 0$. In Lecture 32's polar basis, consider $r(t) = ae^{b\varphi(t)}$, $\varphi(t) = \omega t$, $t \in \mathbb{R}$. Find velocity, speed, acceleration, signed area rate and curvature. Determine exactly when the speed and signed area rate are constant. Use Lecture 8's differentiation rules and Lecture 13's planar cross-product formula. Include $b = 0$ in the conclusions.

Exercise 3 (Curvature and distance along a helix). Let $\boldsymbol{r}(t) = (a\cos t, a\sin t, bt)$, $t \in \mathbb{R}$, where $a > 0$ and $b \in \mathbb{R}$. Find the speed, unit tangent, principal unit normal and curvature. Show explicitly that acceleration is entirely normal. Find the signed arc-length coordinate with base time zero, and check $\kappa = \|\mathrm{d}\boldsymbol{T}/\mathrm{d}s\|$ using that coordinate. Compare the length and the displacement norm over $0 \le t \le 2\pi$. Use Lecture 28's length formula and Lecture 12's norm. Include $b = 0$.

Exercise 4 (A stationary parameter and a geometric tangent). Consider $\boldsymbol{r}(t) = (t^2, t^3)$, $t \in \mathbb{R}$. Find the regular parameter values and the curvature at those values. Compute the one-sided limits of the unit tangent as $t \to 0$ and the limit of curvature as $t \to 0$, $t \neq 0$. Compare these with the limit of $(\boldsymbol{r}(t) - \boldsymbol{r}(0))/\|\boldsymbol{r}(t) - \boldsymbol{r}(0)\|$. Here a geometric tangent line means the line through the origin in the direction of that limiting normalised position secant, when the limit exists. Explain why this line can exist while Lecture 32's unit tangent and curvature are undefined there. Use Lecture 5's one-sided limits and Lecture 12's norm.

Exercise 5 (An ellipse traversed with constant area rate ⋆). Let $p > 0$, $0 \le \varepsilon < 1$ and $\ell > 0$. Suppose a C^2 polar motion on an open interval satisfies

$$r = \frac{p}{1 + \varepsilon \cos \varphi}, \qquad r^2 \dot{\varphi} = \ell.$$

No existence theorem for this branch is assumed or requested. Prove from Lecture 32's polar acceleration identities that

$$\boldsymbol{a} = -\frac{\ell^2}{p r^2} \boldsymbol{e}_r,$$

and find its signed area rate. Include $\varepsilon = 0$. Use the chain rule of Lecture 8, treating the reciprocal radius as a function of angle; do not assume a force law.